

Olympiad Test Paper — Science (Class 7) — Paper 4

Chapter 9: Soil

Paper 4 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	9 — Soil
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Classic traps: swapping percolation rate with water-holding capacity, confusing the A horizon (loses washed-down material) with the B horizon (gains it), assuming 'more water always helps the crop' regardless of aeration, and treating a faster drain time as proof of coarser texture when depth or volume differs. Do not write on this paper — mark answers on your answer sheet.

1. Granite on a humid coastal hill weathers chemically about ten times faster than the same granite on a cold, dry desert plateau, even though both rocks are identical. Two students debate. The soundest single explanation is that the difference is driven by:

- (A) the rock chemistry alone, since identical rock must always weather at one fixed rate regardless of surroundings
- (B) the climate — warmth and abundant moisture both accelerate the chemical breakdown of the rock, so weathering depends on environment as much as on rock type
- (C) the altitude alone, because higher land always weathers rock more rapidly than lower land
- (D) the age of the rock, because older rock is automatically softer and crumbles faster

2. A glacier retreats and exposes bare rock. A team records topsoil depth at 100, 1000 and 10000 years and finds it deepens steadily but very slowly. A visitor concludes 'a good flood could build the same depth of soil in one season.' This conclusion is:

- (A) wrong, because soil deepens only as rock slowly weathers and organisms gradually add humus over very long periods, not in a single flood
- (B) correct, because flowing flood water carries ready-made soil that simply settles in place within days
- (C) correct, because the youngest sites in the data already showed deep soil, proving speed is possible
- (D) wrong, but only because floods remove soil rather than ever adding any depth at all

3. On a single hill the south-facing slope is warm and well-watered and has deep dark soil; the north-facing slope is shaded, cold and dry and has thin pale soil. The same parent rock underlies both. The deeper soil on the warm slope is best explained by:

- (A) the south slope having a completely different and softer bedrock than the north slope
- (B) gravity pulling extra soil downhill specifically onto the south-facing side
- (C) faster weathering plus more plant litter decaying into humus, both favoured by the warmer, moister conditions there
- (D) the warm slope draining water away faster, leaving more minerals behind in the topsoil

4. A student lists agents that turn rock into soil: (i) water freezing in cracks, (ii) roots prying rock apart, (iii) wind-blown sand scouring surfaces, (iv) a single overnight frost forming a finished soil layer. Which item does NOT belong, and why?

- (A) item (i), because freezing water cannot crack rock, it only melts and runs off harmlessly
- (B) item (iv), because no single overnight event can form a finished soil layer; the other three are genuine slow weathering agents
- (C) item (ii), because plant roots are too soft to exert any force on solid rock
- (D) item (iii), because wind carries no particles able to wear down a rock surface

5. Two plots share climate and parent rock. Plot 1 was kept under forest for 200 years; plot 2 was cleared and cropped bare for 80 years. The forest plot's dark surface layer is far thicker. Reading the cause backward from the result, the thicker dark layer is because the forest plot:

- (A) received continuous leaf litter that rotted into humus while erosion was held in check, letting the humus-rich layer build up and stay
- (B) weathered its bedrock faster, and the extra rock fragments are what make the surface layer dark
- (C) had its lower subsoil swell and push the dark layer upward to a greater thickness
- (D) drained rain faster, so dissolved minerals were trapped at the surface and darkened it

6. Identical limestone slabs are placed in four settings for ten years. Ranking how much they break down, the slab that weathers MOST is the one in:

- (A) a warm rainy forest where water, roots and acids from decaying leaves all act on it
- (B) a sealed dry glass case kept at room temperature with no water or air movement
- (C) a freezer held permanently below freezing so no liquid water ever touches it
- (D) a dark dry cupboard with constant air but never any moisture

7. A geologist says soil and bedrock differ in one essential way related to life. The statement that captures this difference correctly is:

- (A) bedrock contains the humus and organisms, while soil is the lifeless rock layer above it
- (B) soil is weathered mineral material mixed with humus and living organisms, whereas bedrock is solid unweathered rock essentially without those living components
- (C) both soil and bedrock are equally full of humus and organisms throughout
- (D) soil is pure organic matter only, with no mineral content from weathered rock at all

8. A pit shows, from the surface down: a crumbly dark band, then a paler firmer band stained with clay and iron that washed down from above, then loose rock chips, then solid rock. The pale stained band is best identified as the:

- (A) A horizon, because the topsoil is always the layer richest in washed-down clay and iron
- (B) C horizon, because partly weathered parent rock is where downward-moving clay first collects
- (C) B horizon, the zone where material leached from the layer above accumulates
- (D) bedrock, because the solid base is where clay and iron from above finally settle

9. In a soil profile, the A horizon is often called a 'zone of loss' and the B horizon a 'zone of gain.' The single process that links the two, transferring fine clay and dissolved minerals from one to the other, is:

- (A) capillary water rising from the bedrock and lifting minerals up into the A horizon
- (B) wind blowing fine clay sideways out of the B horizon into the A horizon
- (C) rainwater seeping downward, washing fine particles and dissolved minerals out of the A horizon and depositing them in the B horizon
- (D) earthworms carrying mineral grains upward from the C horizon to the surface

10. A farmer scrapes off and sells the top 15 cm of his field's soil to a builder. Two years later crop yields have collapsed even though he still ploughs the same depth. The best explanation tied to the profile is that he removed the:

- (A) bedrock, so the field can no longer be weathered into any new soil
- (B) B horizon, which is the only layer that supplies water to crops
- (C) humus-rich A horizon where most nutrients and feeding roots lie, leaving the less fertile subsoil exposed
- (D) C horizon, the topmost and most fertile layer of every profile

11. A claim states: 'The C horizon is simply broken-up bedrock and is barely touched by humus.' Judged against how a profile forms, this claim is:

- (A) wrong, because the C horizon is actually the dark humus-rich topsoil
- (B) broadly correct, because the C horizon is partly weathered parent rock with little organic matter mixed in
- (C) wrong, because the C horizon lies above the A horizon at the surface
- (D) wrong, because the C horizon is made entirely of living organisms and decayed leaves

12. Two profiles are compared. Profile 1 (old grassland): thick dark A, clear B, C, bedrock. Profile 2 (eroded slope): the dark band is missing, and the surface is now a pale firm layer stained with washed-down clay. Profile 2's surface is most likely the exposed:

- (A) C horizon, meaning the bedrock has risen to the surface of the slope
- (B) A horizon, simply a paler version of normal topsoil with extra humus
- (C) bedrock, which has weathered overnight into a stained clay layer
- (D) B horizon, revealed because erosion stripped away the A horizon that once lay on top of it

13. Ranking the horizons by how much living and decayed organic matter they typically hold, from MOST to LEAST, gives:

- (A) bedrock, then C horizon, then B horizon, then A horizon
- (B) A horizon, then B horizon, then C horizon, then bedrock
- (C) B horizon, then A horizon, then bedrock, then C horizon
- (D) C horizon, then bedrock, then A horizon, then B horizon

14. A student must collect two samples: one as humus-rich as possible, one of barely-weathered parent rock fragments. The correct digging plan is:

- (A) take both samples from the B horizon, since the subsoil holds the most of everything
- (B) take the humus-rich sample from the C horizon and the rock fragments from the A horizon
- (C) take both samples from the bedrock, since it is the source of all the layers above
- (D) take the humus-rich sample from the A horizon and the parent-rock-fragment sample from the C horizon

15. A desert profile typically shows almost no dark surface layer, while a rainforest profile shows a deep dark one. Linking horizon to environment, the desert's missing dark layer is best explained because:

- (A) heavy desert rainfall constantly washes the dark layer down into the bedrock
- (B) sparse plant life supplies very little litter, so little humus forms to darken and build the A horizon

- (C) desert sand has no bedrock beneath it, so no profile can develop at all
- (D) wind packs desert grains so tightly that humus is squeezed out of the soil

16. A student heats moist soil in a dry test tube and watches droplets condense on the cool upper glass. A classmate insists this only proves the soil contained air. The first student is right that it proves moisture because:

- (A) heating drove off water vapour from the soil, which then condensed into liquid droplets on the cooler glass
- (B) air released from soil always turns into liquid droplets when it is heated
- (C) the droplets are humus melting and running up the inside of the tube
- (D) the glass itself produces water whenever soil is heated near it

17. To show a dry soil clod contains air, a student drops it into a beaker of water and bubbles stream out. A second clod, pressed hard into a tight pellet first, gives far fewer bubbles. The difference shows that:

- (A) pressing destroyed the soil's water, leaving nothing to make bubbles
- (B) the pressed clod absorbed the air from the surrounding water
- (C) bubbles only appear from soil that contains no air at all
- (D) pressing closed the pore spaces between particles, so there was less trapped air to escape as bubbles

18. A student strongly heats a known mass of garden soil in an open dish until no further change, lets it cool, and finds the mass has dropped and the soil looks paler. The most complete explanation of the mass loss is that heating:

- (A) burned away the sand and silt grains, which is why the soil now weighs less
- (B) drove off the soil's moisture and burned away its humus, both of which had contributed to the original mass
- (C) added oxygen to the soil, and oxygen carries negative weight
- (D) converted the soil into solid bedrock, which is lighter than soil

19. Two equal masses of soil, one clayey and one sandy, are each heated to drive off their water; the clayey sample loses far more mass as water vapour. This is consistent with clayey soil having:

- (A) a higher water-holding capacity, so it retained more moisture before heating
- (B) a higher percolation rate, which is why it held more water than the sand
- (C) larger particles with wide gaps that trap extra water
- (D) no air spaces at all, so the whole sample must have been water

20. A sample is described as roughly 45% mineral grains, 25% air, 25% water and 5% humus by volume after rain. If this same sample is then baked dry in the sun, the component that changes the most is the:

- (A) mineral grains, which evaporate away in strong sunshine
- (B) humus, which boils off entirely whenever soil dries in the sun

- (C) air, which disappears completely as the soil dries out
- (D) water, which largely evaporates, and the space it leaves is taken up by more air

21. A demonstration aims to show soil is 'teeming with life.' Which observation, on its own, gives the strongest direct support?

- (A) the soil feels cool and damp to the touch on a summer morning
- (B) the soil drains a measured volume of water in a fixed number of minutes
- (C) a single gram of garden soil, examined closely, reveals millions of micro-organisms together with worms and insects
- (D) the soil leaves a brown stain when rubbed on white paper

22. Earthworms are called 'farmers' friends.' Tying their behaviour to soil properties, their burrowing most directly improves crops by:

- (A) opening channels that increase aeration and let water percolate more freely to the roots
- (B) compacting the soil so its pores close and it holds firmer
- (C) stripping out all the humus so the soil stays clean and bare
- (D) converting loose sandy soil into solid impermeable clay

23. Three samples are described: P = mostly very small tightly packed particles; Q = mostly large loose grains; R = a balanced blend of large, medium and small particles with plenty of humus. Matching P, Q, R to soil types respectively gives:

- | | |
|--------------------------|--------------------------|
| (A) sandy, clayey, loamy | (B) clayey, sandy, loamy |
| (C) loamy, sandy, clayey | (D) clayey, loamy, sandy |

24. A gardener wishes for one soil that has BOTH the highest water-holding capacity AND the fastest percolation. Why can no ordinary soil satisfy both wishes at once?

- (A) the two are actually the same property, so any soil automatically has both at their maximum
- (B) only sandy soil can hold the most water while also draining the fastest
- (C) high water-holding capacity comes from tiny tightly packed pores, but those same tiny pores slow percolation, so the two properties pull in opposite directions
- (D) fast percolation always forces a soil to retain the most water as well

25. After identical rain, a sandy bed, a loamy bed and a clayey bed are checked over the next day. The set of observations most consistent with their known properties is:

- (A) the sandy bed dried first, the clayey bed stayed soggy longest, and the loamy bed was in between
- (B) the clayey bed dried first because clay drains the fastest of all
- (C) the sandy bed stayed soggy longest because sand holds the most water
- (D) all three beds dried at exactly the same moment regardless of type

26. Aeration — the air available to roots — is generally POOREST in:

- (A) loose sandy soil with large air-filled gaps between grains
- (B) freshly ploughed loamy soil with plenty of open pore spaces
- (C) dry sandy soil on a breezy day with air moving freely through it
- (D) water-logged clayey soil, whose tiny pores stay filled with water and leave no room for air

27. A nursery owner mixes a generous amount of well-rotted compost into a pure sandy bed. The most useful combined effect of the compost is to:

- (A) raise the water-holding capacity and add nutrients while still leaving reasonable drainage
- (B) turn the bed instantly into airless solid clay
- (C) stop the bed from ever letting water percolate again
- (D) remove every living organism so the soil stays sterile

28. A field is freshly ploughed and crumbly in spring; by late summer constant foot traffic has packed it hard. Rain now soaks in far more slowly than it did in spring. The best explanation is that compaction:

- (A) closed up the pore spaces between particles, lowering the percolation rate
- (B) changed the soil's particles into uniformly large sand grains
- (C) added humus that made the water drain faster than before
- (D) enlarged the pores, which is why water now moves slowly

29. Two soils are compared for the life they support: a salt-crusts sandy patch and a humus-rich loamy patch. The loamy patch teems with far more organisms chiefly because it offers:

- (A) the largest particles, which most soil organisms strongly prefer
- (B) the fastest drainage, so organisms are never disturbed by water
- (C) moisture, humus as food, and air pockets — the conditions soil organisms need
- (D) the coldest temperature, since cold soil always supports the most life

30. Sandy soil has the highest percolation rate of common soils. A student concludes 'therefore sandy soil must also have the highest water-holding capacity.' This reasoning fails because percolation rate and water-holding capacity are:

- (A) directly related, so the fastest-draining soil also retains the most water
- (B) exactly the same property, so they must always have equal values
- (C) inversely related — fast-draining sand holds little water, while slow-draining clay holds much
- (D) both at their maximum together only in sandy soil

31. An experiment must be a FAIR test of percolation across two soils. A student pours 300 mL on the sandy column but only 180 mL on the clayey column, times both, and compares. The result is unreliable because:

- (A) the two soils should have been mixed together before pouring any water
- (B) the volume of water poured was not kept the same, so the two percolation measurements cannot be fairly compared
- (C) percolation rate simply cannot be measured for clayey soil at all
- (D) the stopwatch should only be started after all the water has finished draining

32. A column drains 360 mL of water in 8 minutes at a steady rate. Its percolation rate is closest to:

- (A) 2880 mL/min
- (B) 0.022 mL/min
- (C) 352 mL/min
- (D) 45 mL/min

33. Soil A drains 240 mL in 6 minutes; soil B drains 240 mL in 30 minutes. The ratio of their percolation rates (A : B) is:

- (A) 1 : 5
- (B) 6 : 30
- (C) 5 : 1
- (D) 1 : 1

34. Soil P percolates 420 mL in 7 minutes; soil Q percolates 480 mL in 12 minutes. Which has the higher rate, and by how much?

- (A) soil Q, because it let the larger total volume of 480 mL through
- (B) soil Q, because it ran for the longer time of 12 minutes
- (C) they are equal, since both are in the hundreds of mL
- (D) soil P, at 60 mL/min versus soil Q's 40 mL/min — higher by 20 mL/min

35. A soil has a steady percolation rate of 18 mL/min. How long will 270 mL of water take to percolate through it?

- (A) 15 minutes
- (B) 4860 minutes
- (C) 0.067 minutes
- (D) 252 minutes

36. At a steady rate, a soil drains 300 mL in 12 minutes. At the same rate, the volume that would drain in 20 minutes is:

- (A) 500 mL
- (B) 180 mL
- (C) 308 mL
- (D) 720 mL

37. Three equal-section columns are tested with equal water. Sandy drains in 4 min, loamy in 8 min, clayey in 16 min. Listing them from HIGHEST percolation rate to LOWEST gives:

- (A) clayey, then loamy, then sandy
- (B) all three are equal because each drained the same volume
- (C) loamy, then sandy, then clayey
- (D) sandy, then loamy, then clayey

38. Column 1 (deep) drains 600 mL in 12 min; column 2 (the SAME soil, half as deep) drains 600 mL in 6 min. A student concludes column 2's soil is 'sandier.' The flaw is that:

- (A) the deeper column must be sandier, since taking longer proves coarser grains
- (B) both columns hold the same soil, and the shorter time mainly reflects the smaller depth the water had to pass through, not a coarser texture
- (C) soil depth has no effect at all on how long percolation takes
- (D) a faster drain time always proves a larger particle size regardless of setup

39. Soil S drains 540 mL in 6 minutes; soil T drains 540 mL in 18 minutes. By what factor is soil S's percolation rate greater than soil T's?

- (A) 2 times greater
- (B) the same, since the volumes are equal
- (C) 3 times greater
- (D) one-third as great

40. Soil 1 drains 480 mL in 8 min; soil 2 drains 480 mL in 24 min; soil 3 drains 480 mL in 6 min. Labelling them by likely texture (sandy / loamy / clayey) gives:

- (A) soil 3 sandy, soil 1 loamy, soil 2 clayey
- (B) soil 1 sandy, soil 3 loamy, soil 2 clayey
- (C) soil 2 sandy, soil 1 loamy, soil 3 clayey
- (D) all three are the same texture, since each drained 480 mL

41. A field measures 4 mL/min in plot A and 64 mL/min in plot B. A farmer wants paddy (which likes standing water) and groundnut (which likes loose, well-drained ground). The best matching is:

- (A) paddy in plot B, because rice grows best where water drains away the fastest
- (B) groundnut in plot A, because groundnut needs the same standing water as rice
- (C) the same crop in both plots, since percolation rate does not affect crop choice
- (D) paddy in plot A because its slow drainage holds water, and groundnut in plot B because its fast drainage gives loose, airy soil

42. From data alone, a student suspects soil X is loamy and soil Y is clayey. Soil X drains 360 mL in 9 min; soil Y drains 360 mL in 36 min. The sound conclusion is:

- (A) Y drains faster, which is consistent with Y being the loam
- (B) the rates are equal, so the two soils must be identical
- (C) X is the clay because it has the higher percolation rate
- (D) X's rate (40 mL/min) is four times Y's (10 mL/min), consistent with X being the more freely draining loam and Y the slow clay

43. A soil drains 200 mL in the first 5 minutes at a steady rate. Assuming the rate continues, the TOTAL volume drained 13 minutes from the start is:

- (A) 320 mL
- (B) 200 mL
- (C) 520 mL
- (D) 260 mL

44. Two columns of the SAME depth are compared. Column 1 drains 600 mL in 10 min; column 2 drains 600 mL in 15 min. By how many mL/min does column 1's percolation rate exceed column 2's?

(A) 5 mL/min

(B) 20 mL/min

(C) 0 mL/min

(D) 100 mL/min

45. Two adjacent fields each drain 480 mL of water: a well-managed field with added humus takes 8 min, while a compacted over-cultivated field takes 24 min. The humus-improved structure most directly explains why the managed field has the:

(A) lower percolation rate, because humus blocks the pores and slows water

(B) same rate as the compacted field, since both drained 480 mL

(C) higher rate only because its particles are larger grains of sand

(D) higher percolation rate (60 mL/min versus 20 mL/min) because its more open pore spaces let water through faster

46. Water poured on a clayey sample sits on top for a long time before slowly seeping in, while on sandy soil it vanishes almost at once. The clay's behaviour is best explained by:

(A) its large particles leaving wide channels that, oddly, slow the flow

(B) the water being absorbed and chemically destroyed by the clay

(C) its tiny, tightly packed particles leaving only very small pores that resist the downward flow of water

(D) clay having a higher percolation rate than sand despite the slow seeping

47. A farmer with deep, moisture-retaining clayey-loam must choose between wheat/gram and cotton. The sound choice and reason is:

(A) cotton, because cotton needs permanently water-logged clay

(B) wheat or gram, because these crops suit the moisture-holding clayey-loam, while cotton prefers lighter sandy-loam

(C) neither crop, because clayey-loam cannot support any crop

(D) cotton, because clayey-loam drains too fast for wheat or gram

48. A gardener insists 'more water always helps any crop,' and keeps his gram crop on clay permanently flooded. For gram this is harmful chiefly because:

(A) flooding makes the clay drain so fast that the roots dry out

(B) flooding fills the clay's fine pores with water, shutting out air, so the roots are starved of oxygen and rot

(C) the extra water turns the clay into sandy soil over a few days

(D) gram is a water plant that actually needs permanent flooding to thrive

49. A student ranks three soils and finds clay slowest, sand fastest by percolation, then declares 'clay is therefore useless for farming.' The best correction is that:

(A) clay's slow rate means it truly can grow no crop of any kind

(B) the slowest-draining soil always grows the most of every crop

(C) percolation rate has nothing to do with which crop a soil suits

(D) a slow percolation rate is exactly what water-loving crops like paddy need, so clay is valuable for the right crop

50. A farmer needs a vegetable bed that stays moist but never water-logged and is easy to till. Soil M holds water moderately and drains the surplus; soil N holds water strongly and drains very slowly. The better choice and reason is:

- (A) soil N, because the more water a soil holds the better for every crop
- (B) soil N, because very slow drainage guarantees the roots always get air
- (C) soil M, because moderate retention with free surplus drainage avoids both drying out and water-logging
- (D) soil M, because it holds no water at all so it can never be water-logged

51. A new plot is bare, fast-draining sandy-loam. The crop best matched to it, with the right reason, is:

- (A) paddy, because rice grows best where water drains away quickly
- (B) any crop equally, since soil type makes no difference to what grows well
- (C) cotton or lentils, because these tolerate lighter, well-drained sandy-loam
- (D) only crops that require permanently flooded fields

52. A bare ploughed hillside loses far more soil each monsoon than a neighbouring forested slope. The clearest reason the forested slope loses less is that:

- (A) tree roots bind the soil while the canopy and leaf litter break the force of falling rain
- (B) the forest soil is simply too heavy for any rain to move it
- (C) the trees stop every drop of rain from ever reaching the ground
- (D) forests have no topsoil at all for the rain to wash away

53. On a steep slope a farmer cuts level step-like platforms across the hill and also ploughs his furrows along the contour lines rather than up and down. Together these two measures fight erosion mainly by:

- (A) making rainwater evaporate before it has any chance to move the soil
- (B) steepening the slope so that water drains off it even faster
- (C) slowing the downhill rush of water so it soaks in instead of carrying soil away
- (D) deliberately removing all the soil before the rains arrive

54. A dry, treeless plain loses topsoil to blowing dust every summer. The most directly effective control here is to:

- (A) build terraces, which are designed for steep hillsides not flat plains
- (B) dig the field deeper so the wind cannot reach the soil surface
- (C) leave the field bare so the wind has nothing loose to pick up
- (D) plant rows of trees as shelter belts to slow the wind across the field

55. Why is the loss of topsoil to erosion treated as far more serious than losing an equal depth of bedrock?

- (A) topsoil is the humus-rich, nutrient-bearing layer crops feed from, and it takes a very long time to rebuild
- (B) bedrock is the only layer that holds any nutrients for crops

- (C) topsoil grows back completely within a single cropping season
- (D) topsoil plays no role in farming, so losing it is harmless

56. A factory dumps plastic waste and toxic chemicals on nearby fields, and over years crops there fail. This is best classified as:

- (A) soil erosion, because the chemicals wash the topsoil downhill
- (B) weathering, because the chemicals break rock into fresh soil
- (C) healthy humus formation that improves the soil's fertility
- (D) soil pollution, which can poison soil organisms and make the soil unfit for crops

57. A soil-conservation officer tells farmers to cover bare fields between crops with a layer of straw mulch. The chief erosion-fighting benefit of the mulch is that it:

- (A) shields the surface from rain impact and wind and holds moisture, so the topsoil stays in place
- (B) makes the soil so heavy that neither wind nor water can move it
- (C) seals the surface completely so that no rain can ever percolate in
- (D) turns the topsoil into hard bedrock that cannot be eroded

58. A profile shows a very thin dark surface layer over a thick, pale, fast-draining sandy layer; crops there grow poorly and keep drying out. The most likely combined cause is:

- (A) too much humus making the soil permanently water-logged
- (B) little humus in the thin topsoil plus fast percolation that drains water down beyond the reach of the roots
- (C) a thick clay subsoil holding far too much water around the roots
- (D) an unusually high water-holding capacity drying the crops out

59. Two students argue about why water percolates faster down a sandy column than a clayey one of the same depth and width with the same volume poured. The correct physical reason is that:

- (A) sand chemically dissolves in the water, opening up extra channels for it
- (B) clay actively pushes the water back upward against gravity
- (C) sand contains no pore spaces, so water simply slides off its smooth surface
- (D) sand's large grains leave wide pore spaces through which water flows quickly, while clay's tiny grains leave narrow pores that resist flow

60. A student claims weathering and erosion are the same process. The clearest correction, with an example of each, is that:

- (A) weathering carries soil away downhill, while erosion breaks rock apart where it sits
- (B) weathering breaks rock in place into smaller particles, as when frost cracks a boulder, whereas erosion carries away the loosened material, as when monsoon runoff sweeps topsoil off a slope
- (C) both terms mean exactly the same removal of soil by wind and water
- (D) weathering only happens to soil and erosion only happens to solid bedrock

Solutions — Science (Class 7), Chapter 9: Soil — Paper 4

Paper 4 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	11	B	21	C	31	B	41	D	51	C
2	A	12	D	22	A	32	D	42	D	52	A
3	C	13	B	23	B	33	C	43	C	53	C
4	B	14	D	24	C	34	D	44	B	54	D
5	A	15	B	25	A	35	A	45	D	55	A
6	A	16	A	26	D	36	A	46	C	56	D
7	B	17	D	27	A	37	D	47	B	57	A
8	C	18	B	28	A	38	B	48	B	58	B
9	C	19	A	29	C	39	C	49	D	59	D
10	C	20	D	30	C	40	A	50	C	60	B

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (B) Identical rock weathering at different rates proves the rock chemistry cannot be the sole cause, so the fixed-rate idea is refuted by the data itself. Height by itself does not speed weathering — the desert plateau is high yet weathers slowly. Rock age is not the variable here; the two sites share the same rock. What differs is climate: heat and water together drive faster chemical and physical breakdown.

2. (A) The recorded pattern — depth rising slowly over thousands of years — directly contradicts the idea of one-season soil building, so any 'correct' verdict fails. Flood water moves sediment but does not manufacture mature, humus-bearing topsoil overnight. The data show the youngest sites had the least soil, not deep soil. Saying floods only remove soil overstates it; the real point is that genuine soil formation is a long weathering-plus-humus process.

3. (C) Both slopes share the same parent rock, so a 'softer bedrock' claim is ruled out by the setup. Gravity would not selectively favour one compass direction. Faster drainage

would carry minerals away, not concentrate them, so that reasoning is backwards. The genuine driver is the warm, moist microclimate, which speeds weathering and supports vegetation whose litter becomes humus.

4. (B) Freezing water expands and does crack rock (frost wedging), so (i) is valid. Roots growing in cracks exert real prying force, so (ii) is valid. Wind-blown grit abrades surfaces, so (iii) is valid. The odd one is (iv): soil formation is gradual and cannot be finished by one night's frost.

5. (A) A dark thick surface layer signals accumulated humus, which comes from decayed litter, not from pale rock fragments. Subsoil does not physically lift the topsoil upward. Fast drainage carries dissolved minerals down, and minerals are not what make topsoil dark. The forest's steady litter supply plus protection from erosion is what built and preserved the humus-rich layer.

6. (A) A sealed dry case removes water, the key agent, so almost nothing happens. A permanent freezer keeps water solid and largely inert against the surface, limiting chemical attack. A dry cupboard still lacks moisture. The warm rainy forest supplies liquid water, root pressure and leaf-derived acids together — the most aggressive combination.

7. (B) Humus and organisms reside in soil, not in solid bedrock, so swapping them is wrong. Bedrock is not teeming with life, so 'equally full' is false. Soil is not pure organic matter — it is mostly weathered mineral particles plus humus, air and water. The defining contrast is that soil carries the living, organic component that solid bedrock lacks.

8. (C) The A horizon is the dark crumbly topsoil that loses material downward, not the layer that gains it. The C horizon is loose rock chips described separately below the pale band. Bedrock is the solid base, also described separately. The pale band stained with washed-down clay and iron is the B horizon, the zone of accumulation.

9. (C) Upward capillary rise would move material the wrong way and could not explain the A horizon losing while the B gains. Wind acts at the surface, not between buried horizons, and would not selectively enrich the B. Earthworm mixing is real but is not the mechanism that systematically leaches and re-deposits clay. Downward-percolating rain is the agent that strips the A and enriches the B.

10. (C) Bedrock lies deep below 15 cm and is not what a plough or a topsoil sale removes. The B horizon is subsoil below the topsoil, not the surface that was scraped. The C horizon is weathered parent rock far down, not the top or the most fertile layer. Selling the top 15 cm strips the A horizon — the humus-rich, nutrient-bearing topsoil — so yields fall.

11. (B) The dark humus-rich layer is the A horizon, not the C, so that correction is misplaced. The C horizon lies deep, below A and B, not above them. It is not made of organisms or leaves — those concentrate near the surface. The claim is broadly right: the C horizon is partly weathered parent rock with little humus.

12. (D) If the C horizon were at the surface it would show loose weathered rock chips, not a clay-stained accumulation band. A pale layer enriched with washed-down clay is not the humus-rich A horizon, and extra humus would darken it, not pale it. Bedrock does not weather overnight into stained clay. Erosion of the A horizon exposes the B horizon, which matches the described stained surface.

13. (B) Organic matter is concentrated near the surface and dwindles with depth, so any order putting bedrock or the C horizon first is reversed or scrambled. The B horizon gains washed-down minerals but less organic matter than the A. The correct gradient runs A (richest) down through B and C to bedrock (essentially none).

14. (D) The B horizon is neither the richest in humus nor a source of loose parent-rock chips, so both-from-B fails. Reversing the two — C for humus, A for rock — is backwards because humus is in the A and rock fragments are in the C. Solid bedrock yields neither humus nor loose fragments. The A horizon gives the humus sample and the C horizon gives the weathered rock fragments.

15. (B) Deserts are dry, so 'heavy rainfall washing it away' contradicts the setting. A profile can still form over weathered material whether or not bedrock is near the surface, and deserts do have rock beneath. Packing does not squeeze out humus. The real reason is biological: scant vegetation means almost no litter, so very little humus accumulates to form a dark A horizon.

16. (A) Air does not condense into water droplets on heating; only water vapour does, so the air explanation is wrong. Humus does not melt into liquid that climbs glass. Glass does not generate water. The droplets are condensed water vapour driven out of the moist soil, which is exactly what demonstrates the soil held moisture.

17. (D) Bubbles come from trapped air, not from water, so 'destroyed the water' is irrelevant. The clod releasing fewer bubbles is losing trapped air, not absorbing air from the water. Bubbles signal the presence of air, so 'only from soil with no air' is backwards. Compressing the clod collapses the pore spaces, leaving less air to bubble out.

18. (B) Sand and silt are mineral grains that do not burn off at these temperatures, so they are not the lost mass. Oxygen does not have negative weight. Soil does not turn into bedrock by heating. The mass falls and the colour pales because moisture evaporates and the dark humus is burned away.

19. (A) A higher percolation rate means water drains away quickly, which would leave less water held, not more, so that is contradictory. Clay has small particles, not large ones with wide gaps. Clay does contain some air spaces, so it cannot be entirely water. The greater water loss reflects clay's high water-holding capacity.

20. (D) Mineral grains do not evaporate in ordinary sunlight. Humus does not boil away from gentle sun-drying. Air does not vanish as soil dries; in fact it increases. Drying

removes water, and the emptied pore space fills with air, so water is the component that changes most.

21. (C) Coolness and dampness describe moisture, not living organisms. A drainage time measures percolation, a physical property, not life. A brown stain shows colour or humus, not organisms. Finding millions of micro-organisms plus worms and insects in a gram of soil directly demonstrates that soil teems with life.

22. (A) Burrowing loosens soil rather than compacting it, so closed pores is the opposite of the effect. Earthworms add to humus through their casts, not strip it away. They do not transmute sand into clay. Their tunnels open the soil, boosting both air supply and water percolation to roots.

23. (B) Tiny tightly packed particles are clay, so P is clayey, ruling out 'sandy' first. Large loose grains are sand, so Q is sandy. A balanced humus-rich mix is loam, so R is loamy. Only the ordering clayey, sandy, loamy fits all three descriptions.

24. (C) Percolation rate and water-holding capacity are distinct, opposing properties, not one property, so 'the same' is false. Sandy soil drains fast but holds little, so it cannot hold the most water. Fast drainage means water leaves quickly, the reverse of high retention. Small tight pores raise retention but slow drainage, so the two cannot both be maximised.

25. (A) Clay drains slowest and holds the most water, so it cannot dry first. Sand drains fastest and holds the least, so it cannot stay soggiest. Different textures cannot all dry at once. The expected pattern is sandy driest soonest, clayey wettest longest, loamy intermediate.

26. (D) Loose sand has wide gaps full of air, so its aeration is good. Freshly ploughed loam has open pores and good air supply. Dry breezy sand is well aerated. Water-logged clay has fine pores jammed with water, leaving almost no air — the poorest aeration of the four.

27. (A) Compost does not transmute sand into clay, nor make it airless. It does not seal the soil against all percolation; sand still drains reasonably. It feeds organisms rather than sterilising the soil. Its real benefit is boosting water retention and nutrients in the otherwise fast-draining, poor sand while keeping decent drainage.

28. (A) Trampling does not change particle sizes into sand grains. It does not add humus, and added humus would speed not slow infiltration anyway. It does not enlarge pores — enlarged pores would speed drainage, contradicting the observation. Packing collapses the pores, so water percolates more slowly through the hardened soil.

29. (C) Large particles characterise sand, which here is the poorer habitat, so particle size is not the reason. Fast drainage would dry the soil and remove the moisture organisms need. Cold does not generally maximise soil life. The loamy patch wins because it supplies moisture, humus food and air pockets together.

- 30. (C)** If they were directly related, fast drainage would mean high retention, but observation shows the opposite. They are not the same property measured twice. They are not both maximised in sand — sand holds the least. They move oppositely: high percolation goes with low retention, and vice versa.
- 31. (B)** Mixing the soils would destroy the very comparison being made. Clay's percolation rate can be measured; it is just slow. Timing must begin when water is poured, not after draining ends. The flaw is the unequal water volume, which breaks the fair-test rule that only the soil should differ.
- 32. (D)** Percolation rate is volume divided by time: $360 \div 8 = 45$ mL/min. Multiplying 360×8 gives 2880, the wrong operation. Dividing time by volume ($8 \div 360$) gives the inverted 0.022. Subtracting $360 - 8$ gives 352, which mixes unlike quantities. The correct rate is 45 mL/min.
- 33. (C)** Soil A's rate is $240 \div 6 = 40$ mL/min; soil B's is $240 \div 30 = 8$ mL/min, so $A : B = 40 : 8 = 5 : 1$. Writing $1 : 5$ inverts it — A is faster, not slower. Quoting $6 : 30$ confuses times with rates and is even reversed. The volumes are equal but the rates clearly differ, so $1 : 1$ is wrong.
- 34. (D)** Compute rates: $P = 420 \div 7 = 60$ mL/min; $Q = 480 \div 12 = 40$ mL/min. The larger total volume does not mean the higher rate, so Q's 480 mL is a trap. A longer time alone signals slower, not faster, percolation. Being 'in the hundreds' says nothing about rate. P is faster by $60 - 40 = 20$ mL/min.
- 35. (A)** Time equals volume divided by rate: $270 \div 18 = 15$ minutes. Multiplying 270×18 gives 4860, the wrong operation. Dividing rate by volume ($18 \div 270$) gives the inverted 0.067. Subtracting $270 - 18$ gives 252, mixing unlike units. The answer is 15 minutes.
- 36. (A)** The rate is $300 \div 12 = 25$ mL/min, so in 20 minutes $25 \times 20 = 500$ mL drain. 180 mL would mean less water in more time, which is impossible at a steady rate. 308 mL comes from wrongly adding $300 + 8$. 720 mL overshoots by misusing the figures. The correct volume is 500 mL.
- 37. (D)** Equal volume in less time means a higher rate, so the fastest-draining sandy is highest and the slowest clayey is lowest; clayey-first reverses this. Equal volumes do not make equal rates when times differ. Loamy is intermediate, not first. The order from highest to lowest rate is sandy, loamy, clayey.
- 38. (B)** Taking longer does not prove coarser grains — depth was the variable, not texture. Depth clearly does affect drain time, since the half-depth column drained faster. A faster time does not always prove coarser particles when other factors like depth differ. The soils are identical; the time gap comes from depth, not texture.

39. (C) S's rate is $540 \div 6 = 90$ mL/min; T's is $540 \div 18 = 30$ mL/min, so S is $90 \div 30 = 3$ times faster. A factor of 2 misreads the times. Equal volumes do not make equal rates when times differ. 'One-third' inverts the comparison — S is faster, not slower. S's rate is three times T's.

40. (A) Rates are: soil 3 = 80 mL/min (fastest), soil 1 = 60 mL/min (middle), soil 2 = 20 mL/min (slowest). The fastest is the coarsest sandy, the slowest the finest clayey, the middle loamy. Calling soil 1 sandy ignores that soil 3 is faster. Calling soil 2 sandy reverses fastest and slowest. Equal volumes do not equal texture when times differ.

41. (D) Paddy needs water held around the roots, so it suits the slow plot A, not the fast-draining plot B. Groundnut needs drainage and air, the opposite of standing water, so it does not suit slow plot A. Percolation rate strongly affects crop choice. The sound match is paddy in slow A and groundnut in fast B.

42. (D) Y took four times as long for the same volume, so Y is slower, not faster — it cannot be the freer-draining loam. Equal volumes with very different times do not give equal rates. A high percolation rate marks the freely draining loam, not clay, so calling X clay reverses it. X at 40 mL/min is four times Y at 10 mL/min, matching loam over clay.

43. (C) The rate is $200 \div 5 = 40$ mL/min; over the full 13 minutes that is $40 \times 13 = 520$ mL. 320 mL counts only the remaining 8 minutes, not the total. 200 mL ignores the extra time entirely. 260 mL uses the wrong arithmetic. The total after 13 minutes is 520 mL.

44. (B) Column 1's rate is $600 \div 10 = 60$ mL/min; column 2's is $600 \div 15 = 40$ mL/min, a difference of 20 mL/min. 5 mL/min comes from subtracting the times (15 – 10) rather than the rates. 0 mL/min wrongly assumes equal volume means equal rate. 100 mL/min adds the rates instead of subtracting. The excess is 20 mL/min.

45. (D) Humus and loose structure open pores, raising the rate, so 'humus blocks pores' is backwards. Equal volume with very different times does not give equal rates. The difference is from structure, not from the particles suddenly becoming sand. The managed field drains $480 \div 8 = 60$ mL/min against the compacted field's $480 \div 24 = 20$ mL/min.

46. (C) Clay has small particles, not large ones with wide channels — wide channels would speed flow anyway. Water is not chemically destroyed; it is merely slowed. Slow seeping means a low percolation rate, so claiming clay's rate is higher than sand's contradicts the observation. The fine, tightly packed pores of clay resist downward flow, causing the pooling.

47. (B) Cotton favours lighter, freer-draining sandy-loam, not water-logged clay, so cotton is the weaker fit here. Clayey-loam is in fact a productive soil, so 'supports nothing' is false. Clayey-loam holds water and drains slowly, the opposite of draining too fast. Wheat and gram match the moisture-holding clayey-loam.

48. (B) Flooding clay does not speed drainage or dry the roots — clay holds water tightly. Water does not transmute clay into sand. Gram is not a water plant; it does not want permanent flooding. Water-logging fills the pores, excludes air, and suffocates the roots, which is why permanent flooding harms gram.

49. (D) Clay's slow drainage suits paddy and other moisture-loving crops, so it is far from useless. The slowest soil is not best for every crop — groundnut, for instance, needs drainage. Percolation rate is in fact central to crop suitability. Clay's value lies in matching it to the right water-loving crop.

50. (C) Holding the most water is not always best; water-logging harms roots, so soil N is risky for vegetables. Slow drainage tends to fill pores with water and exclude air, the reverse of guaranteeing air. Soil M does hold some water, not none. Soil M's moderate retention plus surplus drainage gives the balanced, well-aerated bed needed.

51. (C) Paddy needs standing water, the opposite of fast-draining sandy-loam, so rice is a poor fit. Soil type does affect crop success, so 'any crop equally' is false. A fast-draining plot cannot stay flooded, so flooding-dependent crops fail. Cotton and lentils suit the lighter, well-drained sandy-loam.

52. (A) Soil weight is not what protects the slope; binding and rain interception do. Canopies slow and break rain but do not stop all of it from reaching the ground. Forests have rich topsoil — that is what they protect. Roots holding the soil and the canopy plus litter cushioning the rain are what reduce erosion.

53. (C) Terraces and contour ploughing do not work by evaporating water. They reduce, not increase, the slope's effective steepness, so 'steepening' is wrong. They protect soil rather than remove it. By slowing runoff they let water infiltrate rather than sweep topsoil downhill.

54. (D) Terraces address water erosion on slopes, not wind on a flat plain. Digging deeper does not shield the exposed surface from wind. Leaving the field bare maximises, not minimises, wind erosion. Shelter belts of trees slow the wind and protect the topsoil, the right measure for a windy plain.

55. (A) Nutrients for crops lie in the topsoil, not the bedrock, so bedrock-as-nutrient-store is wrong. Topsoil does not regrow in one season; it takes centuries. Topsoil is essential to farming, not harmless to lose. Its humus-rich, nutrient-bearing nature plus its very slow rebuilding make its loss so serious.

56. (D) Erosion is the physical removal of topsoil, not contamination by chemicals. Weathering breaks rock into soil and does not describe toxic dumping. Plastic and toxins do not form healthy humus or improve fertility. Adding harmful waste that poisons organisms and ruins crops is soil pollution.

57. (A) Mulch does not work by adding weight to the soil. It does not seal out all rain; water still infiltrates through it. It does not convert soil to bedrock. Mulch cushions the surface from rain and wind and conserves moisture, keeping the topsoil from being washed or blown away.

58. (B) A thin dark layer means scarce humus, the opposite of 'too much humus,' and the soil drains fast rather than water-logging. The subsoil is described as sandy and fast-draining, not thick clay holding water. High water-holding capacity would keep crops moist, not dry them out — a contradiction. The real causes are scant humus and rapid drainage pulling water past the roots.

59. (D) Sand does not dissolve in water. Clay does not push water upward against gravity; it merely slows downward flow. Sand has plenty of pore spaces — indeed wide ones — so 'no pore spaces' is false. The genuine reason is the difference in pore size: wide gaps in sand speed flow, narrow gaps in clay slow it.

60. (B) Carrying soil away is erosion, not weathering, so that pairing is swapped. The two are distinct, not identical: one breaks rock, the other transports the result. Weathering acts on rock in place and erosion can move loosened soil, so confining them to soil-only and bedrock-only is wrong. Weathering fragments rock where it lies; erosion then removes the loosened material.